llmXive Automated Review of Enhancing Train-Free Infinite-Frame Generation for Consistent Long Videos

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper's factual claims are largely supported by the provided data, but there are minor issues regarding the precision of reported percentages and the qualitative description of experimental overhead.

First, in the Methods section (Sec 3.3), the text states that DCE improves subject and background consistency by "4.7% and 2.0%." These figures correspond exactly to the VideoCrafter2 baseline results (S.C. 97.66 vs 92.92; B.C. 96.99 vs 95.01). However, the Wan2.1 baseline shows different improvements (S.C. 96.46 vs 92.67; B.C. 95.50 vs 93.37). Without explicitly stating that these percentages are specific to the VideoCrafter2 configuration, the claim risks being interpreted as a universal metric for the method. The text should clarify the scope of these numbers.

Second, in the Computational Efficiency Analysis (Appendix), the authors describe the memory increase as "moderate" despite Table 4 showing an increase of only 0.10% to 0.66% (approx. 10-66 MiB on a ~10GB baseline). This descriptor overstates the impact. Given the data, "negligible" or "minimal" is a more accurate and defensible characterization.

Finally, while the claim that MIGA "consistently outperforms" baselines in the NarrLV results (Table 2) is numerically true (every MIGA score is higher), the magnitude of improvement varies significantly across TNA settings. The text could be refined to acknowledge this variance to avoid implying a uniform gain across all conditions. These are minor adjustments to ensure the textual claims align perfectly with the quantitative evidence.

Action items.

- [**writing**] Clarify that the '4.7% and 2.0%' consistency gains in Sec 3.3 apply specifically to the VideoCrafter2 baseline, as Wan2.1 gains differ (3.8% and 2.1%).
- [**writing**] Replace 'moderately' with 'negligible' in the memory analysis (App. Comp. Eff.) to accurately reflect the <1% overhead shown in Table 4.
- [**writing**] Qualify the 'consistent outperformance' claim in NarrLV results to acknowledge varying improvement magnitudes across TNA settings and models.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 presents a visual demonstration of the MIGA model's output but lacks a descriptive caption, which is critical for understanding the context of the generated video strips. Additionally, the resolution of the text blocks and axis labels is too low to be legible.

Figure 2

Figure 2 effectively demonstrates the infinite-frame generation capability of MIGA with three distinct video examples. However, the descriptive text blocks are too small to read easily, and the x-axis uses an unexplained infinity symbol.

Figure 3

Figure 3 effectively visualizes the inference framework comparison between FIFO-Diffusion and the proposed TTA mechanism. The schematic grids, color-coded noise levels, and legend are clear, and the caption accurately describes the distinct stages and noise reduction strategies shown.

Figure 4

The figure effectively visualizes the similarity analysis between clean and noisy latents, but the description of the visual anomaly in the caption conflicts with the description in Figure 9, and the legend in the 3D plot is illegible.

Figure 5

The figure itself is clear and effectively illustrates the modeling insight with video frames and corresponding data plots. However, the caption contains formatting artifacts ('%') and is identical to Figure 4's caption, indicating a likely manuscript error.

Figure 6

The figure effectively visualizes the ablation study with clear row labels and bounding boxes, but the caption contains formatting errors ('yellowYellow') and lacks a direct sentence explicitly mapping the rows to the specific method stages.

Figure 7

The figure is visually clear with well-defined axes and legends, but the caption contains a LaTeX syntax error in the variable name and uses an undefined abbreviation ('O.S.') for the metric shown in panel (a).

Figure 8

Figure 8 presents an ablation on Stage 2 steps but misrepresents the x-axis scale and includes an ambiguous, informal annotation that undermines scientific clarity without caption support.

Figure 9

The figure effectively illustrates the DCE framework with clear visual components, but the caption contains a truncation error at the end and a syntax error in the description of the anomaly detection example.

Figure 10

The figure effectively demonstrates temporal consistency across long video sequences, but the caption is insufficient as it does not provide the specific prompts used for the multi-prompt generation case shown.

Figure 11

The figure presents a sequence of video frames but fails to visually highlight the 'bad case' artifacts mentioned in the caption, leaving the reader to guess what constitutes the failure. Additionally, the image lacks internal labels or titles to describe the scene content.

Figure 12

The figure effectively compares the two models visually and numerically, but the descriptive text at the top contradicts the visual content (cymbal vs. Iron Man), and the caption contains a typo ('blueblue').

Action items.

- [**writing**] Figure 1: The caption is explicitly marked as '(no caption)', yet the figure contains complex visual elements (film strips, text blocks, axes) that require a descriptive caption to explain the content and context.
- [**writing**] Figure 1: The x-axis label 'Frame Num' is present, but the axis tick labels are illegible due to low resolution, making it difficult to verify the scale or specific frame counts.
- [**writing**] Figure 1: The text blocks on the left side describing the video prompts are blurry and difficult to read, hindering the viewer's ability to understand the specific inputs used for generation.
- [**writing**] Figure 2: The x-axis label 'Frame Num' is present, but the axis contains a non-standard infinity symbol (∞) next to '1000' without explanation in the caption or axis labels.
- [**writing**] Figure 2: The text blocks describing the video content (Iron Man, sea turtle, Corgi) are extremely small and difficult to read, reducing the figure's clarity.
- [**science**] Figure 4: The caption states the video case contains a 'consistency anomaly,' but

the visual sequence in panel (a) shows a car that is white in the first three frames and black in the fourth, yet the text in Figure 9 describes the anomaly as a car changing from black to white. This contradiction creates confusion regarding the specific anomaly being analyzed.

- [**writing**] Figure 4: The 3D plot in panel (c) has a legend with very small text ('Noise level=10', etc.) that is difficult to read and distinguish from the grid lines, reducing the clarity of the data series mapping.
- [**writing**] Figure 5 caption contains LaTeX-style comment markers ('%') and appears to be a duplicate of Figure 4's description; the text should be cleaned and verified for uniqueness.
- [**writing**] Figure 5 caption is identical to Figure 4's caption, suggesting a copy-paste error in the manuscript that needs correction.
- [**writing**] Figure 6 caption contains raw formatting artifacts ('yellowYellow', 'redRed') that should be cleaned up for professional presentation.
- [**writing**] Figure 6 caption is grammatically incomplete; it defines what the bboxes indicate but does not explicitly state that rows (a)-(d) correspond to the Baseline, Stage 1, Stage 2, and DCE methods respectively, relying on the reader to infer this from the labels.
- [**writing**] Figure 7 caption: The LaTeX syntax for the variable is malformed as `$_adju` (missing backslash and braces); it should be formatted as `$\delta_{adju}$` to match the rendered axis labels.
- [**writing**] Figure 7 caption: The text 'O.S.' is undefined; the caption should explicitly state that 'O.S.' stands for 'Overall score' to match the y-axis label in panel (a).
- [**science**] Figure 8: The x-axis label 'Stage 2 steps' is misleading because the data points (0, 5, 10, 15, 20, 25, 30, 64) are not evenly spaced, yet the visual bar spacing implies a linear progression; the gap between 30 and 64 is visually compressed compared to the actual numerical difference, distorting the trend interpretation.
- [**writing**] Figure 8: The annotation 'Without Stage 1 !!!' with an arrow pointing to the bar at x=64 is informal and lacks precise definition — it's unclear whether this condition applies only to that point or represents a separate experimental setup; the caption does not clarify this critical distinction.
- [**writing**] Figure 9: The caption text is truncated at the end ('achieve h'), likely cutting off the final word (e.g., 'higher consistency').
- [**writing**] Figure 9: The caption contains a syntax error in the parenthetical example ('When anomalies are detected (, the car's...'), missing the condition or symbol before the comma.
- [**writing**] Figure 10: The caption describes a 'multi-prompt controlled generation case' but fails to list the specific text prompts used for each row, making it impossible to verify the model's adherence to the instructions.
- [**writing**] Figure 10: The image contains frame indices (e.g., #0, #50) but lacks explicit row labels or headers to distinguish the different generation conditions or prompts.

- **[science]** Figure 11: The caption claims to illustrate a 'bad case' resulting from migrating to CogVideoX, but the image contains no visual indicators (e.g., red boxes, arrows, or text annotations) to identify the specific artifacts or inconsistencies, making it impossible to verify the claim.
- **[writing]** Figure 11: The image consists of a sequence of frames with frame numbers (#0, #20, etc.) but lacks a descriptive title or axis labels to explain the context of the scene (e.g., 'SUV on mountain road'), relying entirely on the caption for basic context.
- **[writing]** Figure 12: The caption contains a typo 'blueblue' instead of 'blue' when describing the highlighting of superior results.
- **[writing]** Figure 12: The text description at the top of the figure describes a cymbal changing colors, but the visual content shows Iron Man; the text does not match the image.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript exhibits a moderate overuse of domain-specific acronyms and undefined abbreviations that hinder accessibility for non-specialist readers. While the core concepts are sound, the writing frequently assumes the reader possesses prior knowledge of specific method names and metric abbreviations.

First, the primary method name, **MIGA**, is introduced in the caption of Figure 1 and the text of Section 3 without being explicitly defined in the Introduction. The acronym appears before the full name "MIGA" (which seems to be the method name itself, but the text treats it as an acronym without expansion) or a clear definition of what the letters stand for. If "MIGA" is the name, it should be introduced as "We propose MIGA (Method for Infinite-Frame Generation and Alignment, or similar)" or simply "We propose MIGA" if it is a proper noun, but currently, it feels like an undefined acronym. *Correction*: Upon closer inspection, the text uses "MIGA" as a proper noun but never defines what the letters stand for. If it is an acronym, it must be defined. If it is a name, it should be treated as such, but the lack of definition is jarring.

Second, **Table 2** (NarrLV results) relies heavily on the undefined acronym **TNA** (Text-Number-Alignment? Temporal-Noise-Alignment?). The columns are labeled "TNA=2", "TNA=3", etc., with no explanation in the table caption or the surrounding text. This is a significant barrier to understanding the experimental setup.

Third, the results tables (**Table 1**, **Table 3**, **Table 4**) consistently use the abbreviations **S.C.**, **B.C.**, **M.S.**, **T.F.**, and **O.S.** without a legend. While these are defined in the caption of Figure 3, a reader scanning the tables in isolation (or the PDF without scrolling) cannot interpret the data. Standard practice requires defining these in the table caption or the main text immediately preceding the table.

Fourth, specific technical terms like **UniPC** (Section 4.1) and **MMDiT** (Appendix A.2) are used without expansion. While "UniPC" is a known sampler, "MMDiT" is a specific architecture name that should be expanded (e.g., "Multimodal Diffusion Transformer") upon first use to ensure clarity for a broader audience.

Finally, the term **TTS** (Test-Time-Scaling) is used in Appendix A.3. While the expansion is provided, the acronym "TTS" is overwhelmingly associated with "Text-to-Speech" in the broader AI community. Using it for "Test-Time-Scaling" without a strong contextual warning or alternative phrasing risks confusion.

To improve accessibility, the authors should define all acronyms at their first occurrence in the main text, expand metric abbreviations in table captions, and ensure that method names like MIGA are clearly introduced.

Action items.

- [**writing**] The manuscript exhibits a moderate overuse of domain-specific acronyms and undefined abbreviations that hinder accessibility for non-specialist readers. While the core concepts are sound, the writing frequently assumes the reader possesses prior knowledge of specific method names and metric abbreviations. First, the primary method name, MIGA, is introduced in the caption of Figure 1 and the text of Section 3 without being explicitly defined in the Introduction. The acronym appears before the ful

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logical framework for train-free long video generation, but several claims require tighter alignment with the presented data to ensure full logical consistency.

First, the Conclusion states that MIGA "preserves constant memory usage." This claim is logically inconsistent with the data in Table 3 (memory_analysis), which explicitly reports peak memory consumption increasing from 9929 MiB (500 frames) to 9985 MiB (2000 frames). While the authors note this is a "moderate" increase due to intermediate variables, the term "constant" implies no change with respect to video length. To maintain logical rigor, the claim should be revised to reflect "near-constant" or "sub-linear" memory growth, or the definition of "constant" in this context must be explicitly qualified.

Second, the causal attribution of performance gains to the Two-Stage Training-Inference Alignment (TTA) and Dual Consistency Enhancement (DCE) mechanisms relies on the ablation studies in Tables 1 and 4. In Table 1, the baseline (no TTA, no DCE) is compared against the full model. However, the text does not explicitly confirm that the baseline uses the exact same hyperparameters (e.g., $T=64$, $L_{\mathrm{zig}}=4$) as the full MIGA configuration, other than the absence of the specific modules. If the baseline uses different default settings for the underlying FIFO-Diffusion, the performance gap cannot be solely attributed to TTA/DCE. The experimental setup must explicitly state that the baseline is a controlled variant of MIGA with only the proposed modules disabled.

Third, the explanation of the synergy between Stage 1 and Stage 2 in Section 5.2 (“Stage 1 reduces anomalies, Stage 2 suppresses noise”) is not fully supported by Table 4 (tab:ab_2). Table 4 only varies the number of steps in Stage 2 (e) while keeping Stage 1 active (implied by the context of “Stage 2 steps”). It does not provide a row for “Stage 1 only” or a direct comparison isolating Stage 1’s contribution to anomaly reduction. The claim that Stage 1 specifically reduces anomalies is an inference drawn from the text but lacks direct empirical evidence in the provided table, which focuses on the impact of Stage 2 duration. A more granular ablation isolating Stage 1 is necessary to support this specific causal breakdown.

Finally, the discussion on generalizability to MMDiT architectures (Section 4.1) notes that MIGA “struggles” with models like CogVideoX, resulting in “abnormal outputs” (Fig. 4). While this is a valid limitation, the paper does not logically connect this failure to the specific mechanism of TTA or DCE. It is unclear if the failure is due to the noise-span alignment (TTA) or the cross-attention modification (DCE). Clarifying which component causes the incompatibility would strengthen the logical consistency of the limitations section.

Action items.

- **[writing]** The paper presents a logical framework for train-free long video generation, but several claims require tighter alignment with the presented data to ensure full logical consistency. First, the Conclusion states that MIGA “preserves constant memory usage.” This claim is logically inconsistent with the data in Table 3 (memory_analysis), which explicitly reports peak memory consumption increasing from 9929 MiB (500 frames) to 9985 MiB (2000 frames). While the authors note this is a “moderate” incre

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims that slightly overreach the provided evidence or require more precise qualification.

First, the Conclusion states that MIGA yields “state-of-the-art performance.” While MIGA leads among *train-free* methods, the Appendix (Tab:comparison_sf_methods) explicitly compares MIGA against train-based methods like Infinity-RoPE, which achieves an Overall Score (O.S.) of 98.08 compared to MIGA’s 97.82 (VideoCrafter2-based). Claiming “state-of-the-art” without the “train-free” qualifier is an overstatement given the existence of higher-performing trained baselines in the same table. The authors should refine this to “state-of-the-art among train-free methods” to maintain scientific rigor.

Second, the Abstract and Conclusion assert that the method preserves “constant memory usage” and “naturally supports infinite frame generation.” While the memory overhead is minimal, Table 'memory_analysis' in the Appendix demonstrates a measurable increase in peak memory consumption (from 9929 MiB to 9985 MiB) as the frame count scales from 500 to 2000. This contradicts the strict definition of "constant" memory. The claim should be tempered to reflect "bounded" or "negligible" memory growth rather than strictly constant, as the data shows a positive correlation between length and memory usage.

Finally, the Methods section claims DCE improves subject and background consistency by "4.7% and 2.0%." The ablation study in Table 'tab:ab_1' shows the total improvement from the FIFO baseline (95.02 O.S.) to the full MIGA (97.82 O.S.) is 2.8 points. The specific 4.7% figure appears to correspond to the Subject Consistency (S.C.) metric jump (92.92 to 97.66 is ~4.74 points), but the text presents these as general consistency improvements without specifying the metric. This lack of specificity risks misleading the reader about the magnitude of improvement across all consistency dimensions. Clarifying that these percentages apply specifically to Subject Consistency (and potentially Background Consistency if the math holds for that specific metric) is necessary.

Action items.

- [**writing**] The claim that MIGA achieves 'state-of-the-art' performance (Conclusion) is overreaching. Table 1 in the Appendix shows Infinity-RoPE (a train-based method) scoring 98.08 O.S. vs. MIGA's 97.82. The paper must explicitly qualify 'SOTA' to 'train-free SOTA' or acknowledge the gap with trained methods.
- [**writing**] The assertion that MIGA 'naturally supports infinite frame generation' with 'constant memory' (Abstract/Conclusion) is technically imprecise. Table 'memory_analysis' shows memory increasing from 9929 MiB to 9985 MiB as frames grow from 500 to 2000. While the growth is small, it is not constant. The text should reflect 'bounded' or 'sub-linear' growth rather than 'constant'.
- [**writing**] The paper claims DCE improves consistency by '4.7% and 2.0%' (Methods) based on VBench scores. However, the ablation table (Tab:ab_1) shows the jump from FIFO (95.02) to MIGA (97.82) is 2.8 points, not 4.7. The 4.7% figure likely refers to a specific metric (S.C.) or a different baseline comparison not clearly defined in the text, creating ambiguity about the magnitude of the claimed improvement.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a method (MIGA) for generating consistent, infinite-length videos without retraining foundation models. From a safety and ethics perspective, the primary concern is the potential for dual-use. The ability to generate long, temporally consistent videos significantly lowers the barrier for creating high-quality deepfakes, which can be weaponized for disinformation campaigns, non-consensual intimate imagery, and identity fraud.

The current “Impact Statement” (Section 6) is insufficient. It states that “potential societal impacts are deemed non-exceptional,” which is a dangerous underestimation given the specific capability of the model to maintain subject consistency over extended durations (1000+ frames). This specific feature is the exact requirement for realistic deepfake generation. The authors must revise this section to explicitly acknowledge these risks, discuss the potential for misuse, and outline any intended safeguards or recommendations for responsible deployment (e.g., watermarking, detection tools).

Additionally, the “Human Evaluation Experiments” section (Appendix E001) describes a user study involving 8 human annotators. The manuscript currently lacks any mention of Institutional Review Board (IRB) approval, informed consent processes, or how participant data was anonymized and protected. For any research involving human subjects, even in a pairwise comparison setting, standard ethical protocols must be documented. The authors should add a statement confirming that the study was conducted in accordance with ethical guidelines and that informed consent was obtained.

While the paper does not appear to use private or sensitive datasets (relying on public benchmarks and synthetic generation), the lack of explicit ethical consideration for the *output* of the system and the *process* of human evaluation requires attention before publication.

Action items.

- [**writing**] The Impact Statement (Section 6) dismisses societal impacts as ‘non-exceptional’ despite the method enabling infinite, consistent video generation. This capability significantly lowers the barrier for creating deepfakes and disinformation. The statement must be expanded to explicitly address dual-use risks, including identity fraud and misinformation, and propose mitigation strategies.
- [**writing**] The Human Evaluation section (Appendix E001) describes a user study with 8 annotators but lacks details on IRB approval, informed consent procedures, and data privacy protections for the participants. As this involves human subjects, the manuscript must include a statement confirming ethical oversight and compliance with relevant institutional guidelines.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents MIGA, a train-free method for infinite-frame video generation, claiming significant improvements in consistency over FIFO-Diffusion. While the methodological contribution is clear, the scientific evidence supporting the quantitative claims requires strengthening regarding statistical rigor and experimental design.

First, the core results in Tables 1 and 2 (VBench and NarrLV) present point estimates for metrics like Subject Consistency (S.C.) and Overall Score (O.S.) but omit measures of variance (standard deviation) across the evaluation set. In generative modeling, where outputs are stochastic, a single run or an unreported average can be misleading. For instance, the reported gain of +2.03% in O.S. for the TTA mechanism (Sec. 4.2, Tab. 1) lacks context on whether this difference is statistically significant. The authors should report standard deviations over multiple seeds or prompts and include p-values from appropriate statistical tests (e.g., paired t-tests) to validate that the improvements are robust and not artifacts of random sampling.

Second, the human evaluation (Sec. 4.2, Tab. 3) is described as a "large-scale user study" but is based on only 48 prompts and 8 annotators. This sample size is relatively small for establishing strong human preference claims, especially given the high variance in subjective video quality assessment. The manuscript does not report inter-annotator agreement metrics (such as Cohen's kappa or Krippendorff's alpha), which are essential to verify the reliability of the annotations. Without this, the claim that MIGA "consistently outperforms" baselines across all dimensions is not fully supported by the provided evidence.

Finally, the comparison with training-based methods (Tab. 4) raises questions about the fairness of the evaluation protocol. The paper does not explicitly state whether the same set of prompts was used for both the train-free MIGA and the training-based baselines (e.g., Infinity-RoPE, CausVid). If the test sets differ, or if the prompts were selected to favor the specific strengths of MIGA, the comparison could be biased. The authors must clarify the prompt selection strategy and ensure that the evaluation conditions are identical across all compared methods to support the claim of "comparable performance."

In summary, while the proposed method appears effective, the current evidence lacks the statistical depth required to definitively support the magnitude of the claimed improvements. Addressing the variance reporting, statistical significance, and evaluation protocol transparency is necessary before the claims can be considered fully robust.

Action items.

- **[science]** The ablation studies (Tab. 1, Tab. 2) report single-point performance metrics without standard deviations or confidence intervals. Given the stochastic nature of diffusion sampling, statistical significance testing (e.g., paired t-tests) is required to confirm that the observed gains (e.g., +2.03% O.S.) are not due to random variance.
- **[science]** The user study (Tab. 3) relies on a small sample size (48 prompts, 8 annotators) without reporting statistical power analysis or inter-annotator agreement (e.g., Krippendorff's alpha). The claim of 'large-scale' validation is unsupported by these numbers; a more rigorous statistical treatment or larger sample is needed.
- **[science]** The comparison with training-based methods (Tab. 4) lacks a clear definition of the evaluation subset. If the same prompts were used for both train-free and train-based methods, potential data leakage or prompt bias could inflate the train-free scores. The experimental protocol must explicitly state how the test set was constructed to ensure fair comparison.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The manuscript presents a compelling method (MIGA) for train-free long video generation, but the statistical rigor of the experimental evaluation requires strengthening to fully support the claims of superiority and stability.

First, the human evaluation results presented in Table 1 (Appendix D) are reported as precise percentages (e.g., 62.23% preference for MIGA) derived from a sample of 48 prompts and 8 annotators. While the point estimates favor the proposed method, the manuscript lacks any measure of statistical significance. Given the sample size, it is unclear if these differences are statistically distinguishable from chance. The authors should perform a binomial test or report 95% confidence intervals for these preference rates to validate that the observed improvements are robust.

Second, the ablation studies (Tables 2 and 3 in the main text and Appendix) report single-point performance metrics for various hyperparameters (e.g., L_{zig}, δ_{adju}). In stochastic generative models, a single run per configuration is often insufficient to distinguish true performance gains from random noise. The absence of standard deviations, error bars, or results from multiple random seeds makes it difficult to assess the stability of the method. For instance, the reported memory increase of 0.10% (Table 4) is extremely small and likely within the margin of error for a single measurement; reporting variance would clarify if this is a consistent overhead or a measurement artifact.

Finally, while the VBench and NarrLV scores are presented as definitive improvements, the manuscript does not discuss the variance inherent in these automated benchmarks. Video generation is inherently stochastic; without reporting results over multiple seeds or providing confidence intervals for the benchmark scores, the claim of "state-of-the-art" performance rests on point estimates that may not be reproducible. The authors should either report mean and standard deviation over multiple runs or explicitly state the limitations of single-run reporting in the context of their conclusions.

Action items.

- [**science**] The user study (Tab. 1, App. D) reports percentages (e.g., 62.23%) from 48 prompts and 8 annotators but lacks confidence intervals or significance tests (e.g., binomial test) to confirm the improvements over FIFO-Diffusion are statistically significant rather than random variation.
- [**science**] Ablation tables (Tab. 2, Tab. 3) present single-point performance metrics without standard deviations or error bars, making it impossible to assess the stability of the reported gains (e.g., the 0.01% memory increase) or the robustness of hyperparameter choices.
- [**science**] The claim of 'state-of-the-art' performance relies on point estimates from VBench and NarrLV; the manuscript should clarify if these benchmarks provide variance estimates

or if the authors performed multiple runs to ensure the observed differences are not due to stochasticity in the generation process.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high standard of technical writing, with a clear logical flow from the introduction of the problem to the proposed methodology and experimental validation. The abstract and introduction effectively set the stage, and the transition between the "Methods" and "Experiments" sections is smooth. The use of active voice in describing the proposed MIGA framework (e.g., "MIGA augments...") enhances readability.

However, there are a few minor grammatical and stylistic issues that should be addressed to polish the final version. In the appendix section "Implementation on Different Foundation Models," the sentence "However, We observe that this frame-level autoregressive generation framework..." contains a capitalization error; "We" should be lowercase. Additionally, in the "Acknowledgements" section, the citation of the grant number lacks a standard space between "No." and the number ("Grant No.2022ZD0116403" should be "Grant No. 2022ZD0116403").

The description of the limitations in the final section is generally clear, though the phrasing regarding the "cat's head and tail" switching places could be slightly refined to ensure the nature of the hallucination is unambiguous to the reader. Overall, the paper is well-structured and the prose is professional, requiring only these minor corrections before publication.

Action items.

- **[writing]** In the 'Implementation on Different Foundation Models' section, the sentence 'However, We observe that...' contains a capitalization error. The pronoun 'We' should be lowercase ('we') as it follows a comma and is not the start of a new sentence.
- **[writing]** The 'Acknowledgements' section lists 'Grant No.2022ZD0116403'. Standard English convention requires a space between the abbreviation 'No.' and the number (i.e., 'No. 2022ZD0116403').
- **[writing]** In the 'Limitations and Future Work' section, the phrase 'cat's head and tail suddenly switch places' is slightly ambiguous. Consider clarifying if this refers to a spatial swap or a morphological transformation to ensure precise reader understanding.